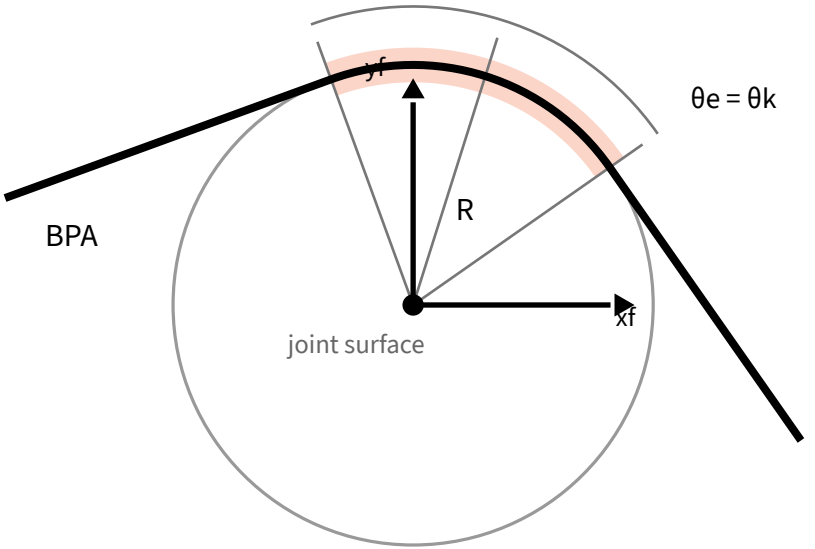
$\phi = |\theta wrap - \theta k|$

$\theta s = \theta wrap$ (contact begins)



shaded arc $s = R \cdot \phi$: usable length lost to wrapping

femur frame {f}